Chapter 3. Inference for Stationary ARMA Models

Estimation of mean and ACF, Yule-Walker, LS, ML, prediction

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3.1 Estimation of the mean

Motivation. The CLT for $\bar{X}$ survives dependence, but with an asymptotic variance that depends on the model and can differ greatly from $\mathrm{Var}(X_t)$.

The subscript n is dropped ($\bar{X} = \bar{X}_n$) and limits are as $n \to \infty$. Consider the AR(1) model

$$X_t - \mu = \phi(X_{t-1} - \mu) + \epsilon_t, \qquad \{\epsilon_t\} \sim IID(0, \sigma^2), \quad |\phi| < 1.$$

By (2.5),

$$X_t = \mu + \sum_{j=0}^{\infty} \phi^j \epsilon_{t-j}. \tag{3.1}$$

Theorem 3.1.1 (CLT for the sample mean of an AR(1))

If $\{X_t\}$ is the stationary series (3.1) and $\hat{\mu} = \bar{X} = n^{-1}\sum_{t=1}^{n} X_t$, then

$$\sqrt{n}(\hat{\mu} - \mu) \xrightarrow{d} N\Big(0, \frac{\sigma^2}{(1-\phi)^2}\Big).$$

Proof.

Step 1. (Decomposition.) Put $b_j = \sum_{k \geq j+1} \phi^k = \phi^{j+1}/(1-\phi)$ and $y_t = \sum_{j \geq 0} b_j \epsilon_{t-j}$. Since $\sum |b_j| < \infty$, $\{y_t\}$ is stationary (Lemma 2.2.A). As $b_{j-1} - b_j = \phi^j$ and $b_0 = \phi/(1-\phi)$,

$$\begin{aligned} X_t - \mu &= \epsilon_t + \sum_{j \geq 1} (b_{j-1} - b_j)\epsilon_{t-j} = \Big(\sum_{j \geq 0} \phi^j\Big)\epsilon_t - b_0\epsilon_t + \sum_{j \geq 0} b_j \epsilon_{t-1-j} - \sum_{j \geq 1} b_j \epsilon_{t-j} \\ &= \Big(\sum_{j \geq 0} \phi^j\Big)\epsilon_t - y_t + y_{t-1}. \end{aligned}$$

Step 2. (Telescoping.) Summing over t,

$$\sqrt{n}(\hat{\mu} - \mu) = \frac{1}{\sqrt{n}} \sum_{t=1}^{n} (X_t - \mu) = \underbrace{\frac{1}{1-\phi} \cdot \frac{1}{\sqrt{n}} \sum_{t=1}^{n} \epsilon_t}_{\text{(I)}} \underbrace{- \frac{1}{\sqrt{n}} y_n + \frac{1}{\sqrt{n}} y_0}_{\text{(II)}}.$$

Step 3. By the i.i.d. CLT, (I) $\xrightarrow{d} N(0, \sigma^2/(1-\phi)^2)$, since $\sum_j \phi^j = 1/(1-\phi)$.

Step 4. $E(\text{II})^2 \leq \frac{2}{n}(Ey_n^2 + Ey_0^2) = \frac{4}{n} Ey_0^2 \to 0$, so (II) $\xrightarrow{p} 0$ by Chebyshev's inequality. Slutsky's theorem gives the claim. ∎

Remark 3.1.2 (General linear processes, ARMA)

For $X_t = \mu + \sum_{j \geq 0} a_j \epsilon_{t-j}$, $\{\epsilon_t\} \sim IID(0, \sigma^2)$, $|a_j| \leq c\rho^j$ ($0 < \rho < 1$; every causal ARMA by Proposition 2.4.A), put $b_j = \sum_{k \geq j+1} a_k$, $|b_j| \leq c\rho^{j+1}/(1-\rho)$, $y_t = \sum_j b_j \epsilon_{t-j}$. As above $X_t - \mu = (\sum_j a_j)\epsilon_t - y_t + y_{t-1}$, so

$$\sqrt{n}(\hat{\mu} - \mu) = \frac{1}{\sqrt{n}}\Big(\sum_{j=0}^{\infty} a_j\Big) \sum_{t=1}^{n} \epsilon_t - \frac{1}{\sqrt{n}} y_n + \frac{1}{\sqrt{n}} y_0 \xrightarrow{d} N\Big(0, \Big(\sum_{j=0}^{\infty} a_j\Big)^2 \sigma^2\Big).$$

To test $H_0 : \mu = 0$ at level α, reject when $|\sqrt{n}\bar{X}| \geq z_{\alpha/2} |\sum_j a_j| \sigma$ (i.i.d.: $z_{\alpha/2}\sigma$), where $P(|Z| \leq z_{\alpha/2}) = 1 - \alpha$. For AR(1), $\sum_j a_j = 1/(1-\phi)$ is huge for $\phi \approx 1$; the $100(1-\alpha)\%$ interval is

$$\bar{X} - \frac{z_{\alpha/2}\sigma}{\sqrt{n}(1-\phi)} \leq \mu \leq \bar{X} + \frac{z_{\alpha/2}\sigma}{\sqrt{n}(1-\phi)}.$$

3.2 Sample autocovariance, autocorrelation and partial autocorrelation

Motivation. Bartlett's theorem tells how far a sample autocorrelation can stray from its true value by chance.

Definition 3.2.1 (Sample autocovariance function)

For $|h| < n$, the *sample autocovariance function* is

$$\hat{\gamma}(h) = \frac{1}{n}\sum_{t=1}^{n-|h|}(X_t - \bar{X})(X_{t+|h|} - \bar{X}).$$

For stationary ARMA series $\hat{\gamma}(h) \to \gamma(h)$ a.s. (ergodic theorem). It is reliable only for small lags (rule of thumb $|h| \le \sqrt{n}$); the divisor n makes $[\hat{\gamma}(i-j)]$ nonnegative definite.

Definition 3.2.2 (Sample autocorrelation function)

The *sample autocorrelation function* (SACF), the estimator of $\rho(h)$, is

$$\hat{\rho}(h) = \frac{\hat{\gamma}(h)}{\hat{\gamma}(0)}.$$

Like ρ, it satisfies $|\hat{\rho}(h)| \le 1$.

Definition 3.2.3 (Sample partial autocorrelation)

For fixed h, the last component $\hat{\phi}_{hh}$ of the solution of

$$\begin{pmatrix} \hat{\rho}(0) & \hat{\rho}(1) & \cdots & \hat{\rho}(h-1) \\ \hat{\rho}(1) & \hat{\rho}(0) & \cdots & \hat{\rho}(h-2) \\ \vdots & \vdots & \ddots & \vdots \\ \hat{\rho}(h-1) & \hat{\rho}(h-2) & \cdots & \hat{\rho}(0) \end{pmatrix} \begin{pmatrix} \hat{\phi}_{h1} \\ \hat{\phi}_{h2} \\ \vdots \\ \hat{\phi}_{hh} \end{pmatrix} = \begin{pmatrix} \hat{\rho}(1) \\ \hat{\rho}(2) \\ \vdots \\ \hat{\rho}(h) \end{pmatrix}$$

is the *sample partial autocorrelation coefficient*; as a function of h it is the *sample partial autocorrelation function* (SPACF).

Theorem 3.2.4 (Bartlett)

Let $\{X_t\}$ be stationary with

$$X_t = \mu + \sum_{j=0}^{\infty} a_j \epsilon_{t-j}, \qquad \{\epsilon_t\} \sim IID(0, \sigma^2), \quad \sum_{j=0}^{\infty} |a_j| < \infty, \quad E\epsilon_t^4 < \infty.$$

Then for each fixed $h \ge 1$,

$$\sqrt{n}(\hat{\boldsymbol{\rho}}(h) - \boldsymbol{\rho}(h)) \xrightarrow{d} N(0, W), \qquad \hat{\boldsymbol{\rho}}(h) = (\hat{\rho}(1), \ldots, \hat{\rho}(h))^T, \quad \boldsymbol{\rho}(h) = (\rho(1), \ldots, \rho(h))^T,$$

where $W = (w_{ij})_{i,j=1}^{h}$ is given by *Bartlett's formula*

$$w_{ij} = \sum_{k=-\infty}^{\infty} \Big\{ \rho(k+i)\rho(k+j) + \rho(k-i)\rho(k+j) + 2\rho(i)\rho(j)\rho^2(k)$$

$$- 2\rho(i)\rho(k)\rho(k+j) - 2\rho(j)\rho(k)\rho(k+i)\Big\}$$
$$= \sum_{k=1}^{\infty}\{\rho(k+i)+\rho(k-i)-2\rho(i)\rho(k)\}\{\rho(k+j)+\rho(k-j)-2\rho(j)\rho(k)\}.$$

Proof. [Supplement] Two standard limit theorems are quoted. Write $\eta = E\epsilon_t^4/\sigma^4$ and, without loss of generality, $\mu = 0$. Let $\gamma^*(h) = n^{-1}\sum_{t=1}^{n} X_t X_{t+h}$.

Step 1. (Fourth moments.) Since $E\epsilon_i\epsilon_j\epsilon_k\epsilon_l = \sigma^4(\delta_{ij}\delta_{kl}+\delta_{ik}\delta_{jl}+\delta_{il}\delta_{jk}) + (\eta-3)\sigma^4\delta_{ijkl}$, expanding the four series ($a_j = 0$ for $j<0$) gives

$$E[X_tX_{t+p}X_sX_{s+q}] = \gamma(p)\gamma(q)+\gamma(s-t)\gamma(s-t+q-p)+\gamma(s-t+q)\gamma(s-t-p)$$
$$+ (\eta-3)\sigma^4\sum_i a_i a_{i+p}a_{i+s-t}a_{i+s-t+q}.$$

Step 2. (Covariances.) Summing over $s,t = 1,\dots,n$ and grouping by $k = s-t$,

$$n\,\mathrm{Cov}(\gamma^*(p),\gamma^*(q)) = \sum_{|k|<n}\Big(1-\frac{|k|}{n}\Big)\Big[\gamma(k)\gamma(k+q-p)+\gamma(k+q)\gamma(k-p)$$
$$+ (\eta-3)\sigma^4\sum_i a_i a_{i+p}a_{i+k}a_{i+k+q}\Big].$$

All terms are absolutely summable in k (as $\sum|\gamma(k)| \le \sigma^2(\sum|a_j|)^2$), so by dominated convergence the limit is

$$V_{pq} = (\eta-3)\gamma(p)\gamma(q) + \sum_{k=-\infty}^{\infty}[\gamma(k)\gamma(k+q-p)+\gamma(k+q)\gamma(k-p)],$$

using $\sum_k\sum_i a_i a_{i+p}a_{i+k}a_{i+k+q} = \sigma^{-4}\gamma(p)\gamma(q)$.

Step 3. (Joint CLT for γ^*.) For the truncated process $X_t^{(m)} = \sum_{j=0}^{m} a_j\epsilon_{t-j}$, the vectors $(X_t^{(m)}X_{t+l}^{(m)})_{l=0}^{h}$ form a strictly stationary $(m+h)$-dependent sequence, so the CLT for m-dependent sequences (Hoeffding–Robbins) together with the Cramér–Wold device gives $\sqrt{n}(\boldsymbol{\gamma}_m^* - \boldsymbol{\gamma}_m) \xrightarrow{d} N(0, V^{(m)})$, where $V^{(m)} \to V$ as $m\to\infty$ by Step 2 applied to $a^{(m)}$. Applying the computation of Step 2 to the difference $X_t - X_t^{(m)}$ gives $\limsup_n n\,\mathrm{Var}(\gamma^*(l)-\gamma_m^*(l)) \le C\sum_{j>m}|a_j| \to 0$ as $m\to\infty$. By the standard approximation lemma (Billingsley, Thm. 3.2), $\sqrt{n}(\boldsymbol{\gamma}^* - \boldsymbol{\gamma}) \xrightarrow{d} N(0,V)$, with $\boldsymbol{\gamma}^* = (\gamma^*(0),\dots,\gamma^*(h))^T$.

Step 4. (From γ^* to $\hat\gamma$.) Expanding the product in Definition 3.2.1,

$$\hat\gamma(l) - \gamma^*(l) = -\frac{1}{n}\sum_{t=n-l+1}^{n} X_tX_{t+l} - \bar X\cdot\frac{1}{n}\sum_{t=1}^{n-l}(X_t+X_{t+l}) + \Big(1-\frac{l}{n}\Big)\bar X^2.$$

The first term has L^1 norm $\le l\gamma(0)/n$. The other two are bounded by $C\bar X^2 + O_p(n^{-1})|\bar X|$, and $\bar X = O_p(n^{-1/2})$ by Remark 3.1.2. Hence $\sqrt{n}(\hat{\boldsymbol{\gamma}} - \boldsymbol{\gamma}^*) \xrightarrow{p} 0$ and $\sqrt{n}(\hat{\boldsymbol{\gamma}} - \boldsymbol{\gamma}) \xrightarrow{d} N(0,V)$.

Step 5. (Delta method.) $\hat\rho(i) = g_i(\hat{\boldsymbol{\gamma}})$ with $g_i(x) = x_i/x_0$, whose gradient at $\boldsymbol{\gamma}$ has entries $-\rho(i)/\gamma(0)$ (in x_0) and $1/\gamma(0)$ (in x_i). Hence $\sqrt{n}(\hat{\boldsymbol{\rho}}(h) - \boldsymbol{\rho}(h)) \xrightarrow{d} N(0,W)$ with

$$w_{ij} = \frac{V_{ij} - \rho(i)V_{0j} - \rho(j)V_{i0} + \rho(i)\rho(j)V_{00}}{\gamma(0)^2}.$$

The $(\eta-3)$ terms contribute $(\eta-3)[\rho(i)\rho(j)-\rho(i)\rho(j)-\rho(j)\rho(i)+\rho(i)\rho(j)] = 0$, so W does not depend on η. With $T_{pq} = V_{pq}/\gamma(0)^2$ (without the η term) and the shifts $k \mapsto k+i$, one gets

$T_{ij} = \sum_k [\rho(k+i)\rho(k+j) + \rho(k-i)\rho(k+j)]$, $T_{0j} = 2\sum_k \rho(k)\rho(k+j)$, $T_{i0} = 2\sum_k \rho(k)\rho(k+i)$ and $T_{00} = 2\sum_k \rho^2(k)$, which yields the first form of w_{ij}.

Step 6. (Second form.) Expand the product in the second form, sum over $k \geq 1$, and use $\rho(-k) = \rho(k)$ to rewrite each one-sided sum as half of a two-sided sum (the $k = 0$ terms cancel); this reproduces the first form. ■

Corollary 3.2.A (White-noise and MA(q) bands; supplement)

(1) If $\{X_t\} \sim IID(0, \sigma^2)$ with $EX_t^4 < \infty$, then $\sqrt{n}\,\hat{\boldsymbol{\rho}}(h) \xrightarrow{d} N(0, I_h)$.
(2) If $\{X_t\}$ is MA(q), then for $i > q$, $\sqrt{n}\,\hat{\rho}(i) \xrightarrow{d} N\big(0,\, 1 + 2\sum_{k=1}^{q} \rho^2(k)\big)$.

Proof. [Supplement]

Step 1. For i.i.d. data $\rho(k) = 0$ for $k \neq 0$. In the second form of Bartlett's formula, the factor $\rho(k+i) + \rho(k-i) - 2\rho(i)\rho(k)$ with $k \geq 1$, $i \geq 1$ is nonzero only for $k = i$, where it equals 1. Thus $w_{ij} = \mathbb{1}\{i = j\}$.

Step 2. If $i > q$ then $\rho(i) = 0$ and $\rho(k+i) = 0$ for $k \geq 1$, so $w_{ii} = \sum_{k\geq 1} \rho^2(k-i) = \sum_{m=1-i}^{\infty} \rho^2(m) = \rho^2(0) + 2\sum_{k=1}^{q} \rho^2(k)$, since $\rho(m) = 0$ for $|m| > q$. ■

For white noise about 95% of the $\hat{\rho}(h)$ lie within $\pm 1.96/\sqrt{n}$; for an MA(q) the bands at lags $> q$ widen by $(1 + 2\sum_{k\leq q} \rho^2(k))^{1/2}$.

3.3 The method of moments

Motivation. Equating population and sample autocorrelations gives, for AR models, the linear Yule–Walker system.

The moment equations are nonlinear for MA/ARMA, so consider the causal AR(p) model

$$X_t = \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + \epsilon_t, \qquad \{\epsilon_t\} \sim IID(0, \sigma^2).$$

By causality,

$$E\big[(X_t - \phi_1 X_{t-1} - \cdots - \phi_p X_{t-p}) X_{t-j}\big] = E\epsilon_t X_{t-j} = 0, \qquad j = 1, \ldots, p,$$

which gives the *Yule–Walker equations* $\gamma(j) = \phi_1\gamma(j-1) + \cdots + \phi_p\gamma(j-p)$, $j = 1, \ldots, p$; in matrix form

$$\begin{pmatrix} \rho(0) & \rho(1) & \cdots & \rho(p-1) \\ \rho(1) & \rho(0) & \cdots & \rho(p-2) \\ \vdots & \vdots & \ddots & \vdots \\ \rho(p-1) & \rho(p-2) & \cdots & \rho(0) \end{pmatrix} \begin{pmatrix} \phi_1 \\ \phi_2 \\ \vdots \\ \phi_p \end{pmatrix} = \begin{pmatrix} \rho(1) \\ \rho(2) \\ \vdots \\ \rho(p) \end{pmatrix}. \tag{3.2}$$

Replacing $\rho(j)$ by $\hat{\rho}(j)$ gives the *Yule–Walker (Y-W) estimator* $\hat{\boldsymbol{\phi}}_{YW}$; by Definition 3.2.3, the last component of the order-h Y-W estimator is $\hat{\phi}_{hh}$.

Theorem 3.3.1 (Asymptotic normality of the Y-W estimator)

Let $\hat{\boldsymbol{\phi}}_{YW} = (\hat{\phi}_1, \ldots, \hat{\phi}_p)^T$ be the Y-W estimator of $\boldsymbol{\phi}$ in a causal AR(p) model. Then

$$\sqrt{n}(\hat{\boldsymbol{\phi}}_{YW} - \boldsymbol{\phi}) \xrightarrow{d} N(0, \Gamma^{-1}\sigma^2),$$

where Γ is the $p \times p$ matrix with $\Gamma_{ij} = \gamma(i-j)$.

Lemma 3.3.A (Basic limits for causal AR(p); supplement)

Let $\{X_t\}$ be a causal AR(p) with $\{\epsilon_t\} \sim IID(0, \sigma^2)$, and $\mathbf{X}_{t-1} = (X_{t-1}, \dots, X_{t-p})^T$. Then
(a) $\frac{1}{n}\sum_{t=p+1}^{n} \mathbf{X}_{t-1}\mathbf{X}_{t-1}^T \to \Gamma$ a.s., and $\hat{\gamma}(h) \to \gamma(h)$ a.s. for each h;
(b) $\frac{1}{\sqrt{n}}\sum_{t=p+1}^{n} \mathbf{X}_{t-1}\epsilon_t \xrightarrow{d} N(0, \sigma^2\Gamma)$;
(c) Γ is nonsingular.

Proof. [Supplement]

Step 1. (a) $X_t = \sum_j a_j\epsilon_{t-j}$ is a fixed function of the shifted i.i.d. sequence, so $\{X_t\}$ and $X_{t-i}X_{t-j}$ are strictly stationary and ergodic; Birkhoff gives $n^{-1}\sum X_{t-i}X_{t-j} \to \gamma(i-j)$, $\bar{X} \to 0$ and $\hat{\gamma}(h) \to \gamma(h)$ a.s.

Step 2. (b) Let $\mathcal{F}_t = \sigma(\epsilon_s : s \le t)$, $c \in \mathbb{R}^p$, $\xi_t = c^T\mathbf{X}_{t-1}\epsilon_t$: stationary ergodic, $E(\xi_t \mid \mathcal{F}_{t-1}) = c^T\mathbf{X}_{t-1}E\epsilon_t = 0$, $E\xi_t^2 = \sigma^2 c^T\Gamma c$. The martingale-difference CLT (Billingsley–Ibragimov) gives $n^{-1/2}\sum \xi_t \xrightarrow{d} N(0, \sigma^2 c^T\Gamma c)$; apply Cramér–Wold.

Step 3. (c) Step 1 of Proposition 2.5.B: $c^T\Gamma c \ge c_i^2\sigma^2 > 0$ for $c \ne 0$. ■

Proof of Theorem 3.3.1. [Supplement] Let $\hat{\Gamma} = [\hat{\gamma}(i-j)]_{i,j=1}^{p}$ and $\hat{\boldsymbol{\gamma}}_p = (\hat{\gamma}(1), \dots, \hat{\gamma}(p))^T$, so $\hat{\Gamma}\hat{\boldsymbol{\phi}}_{YW} = \hat{\boldsymbol{\gamma}}_p$ (multiply (3.2) with hats by $\hat{\gamma}(0)$).

Step 1. Then $\hat{\Gamma}(\hat{\boldsymbol{\phi}}_{YW} - \boldsymbol{\phi}) = \hat{\boldsymbol{\gamma}}_p - \hat{\Gamma}\boldsymbol{\phi}$, whose j-th entry is $\hat{\gamma}(j) - \sum_k \phi_k\hat{\gamma}(j-k)$.

Step 2. Replacing $\hat{\gamma}(l)$ by $n^{-1}\sum_t X_tX_{t+l}$ costs $O(p)$ boundary terms $O_p(1/n)$ and terms $\bar{X} \cdot n^{-1}\sum X_t = O_p(1/n)$ (Remark 3.1.2), negligible after $\times\sqrt{n}$:

$$\sqrt{n}\Big(\hat{\gamma}(j) - \sum_k \phi_k\hat{\gamma}(j-k)\Big) = \frac{1}{\sqrt{n}}\sum_{t=p+1}^{n} X_{t-j}\Big(X_t - \sum_k \phi_k X_{t-k}\Big) + o_p(1) = \frac{1}{\sqrt{n}}\sum_{t=p+1}^{n} X_{t-j}\epsilon_t + o_p(1).$$

Step 3. By the Lemma, $\sqrt{n}(\hat{\boldsymbol{\gamma}}_p - \hat{\Gamma}\boldsymbol{\phi}) \xrightarrow{d} N(0, \sigma^2\Gamma)$ and $\hat{\Gamma} \to \Gamma$ (nonsingular). By Slutsky,

$$\sqrt{n}(\hat{\boldsymbol{\phi}}_{YW} - \boldsymbol{\phi}) = \hat{\Gamma}^{-1}\sqrt{n}(\hat{\boldsymbol{\gamma}}_p - \hat{\Gamma}\boldsymbol{\phi}) \xrightarrow{d} N(0, \Gamma^{-1}\sigma^2\Gamma\Gamma^{-1}) = N(0, \Gamma^{-1}\sigma^2).$$

■

Example 3.3.3 (Y-W estimators for AR(1) and AR(2))

For AR(1), $\rho(1) = \phi$ gives $\hat{\phi} = \hat{\rho}(1)$. For AR(2) the Y-W equations are

$$\rho(1) = \phi_1 + \rho(1)\phi_2, \qquad \rho(2) = \rho(1)\phi_1 + \phi_2,$$

so

$$\hat{\phi}_1 = \frac{\hat{\rho}(1)(1-\hat{\rho}(2))}{1-\hat{\rho}^2(1)}, \qquad \hat{\phi}_2 = \frac{\hat{\rho}(2)-\hat{\rho}^2(1)}{1-\hat{\rho}^2(1)}.$$

Corollary 3.3.B (Sample PACF of an AR(p); supplement)

For a causal AR(p) and each $h > p$, $\sqrt{n}\,\hat{\phi}_{hh} \xrightarrow{d} N(0,1)$.

Proof. [Supplement]

Step 1. An AR(p) is also an AR(h) with $\boldsymbol{\phi}^{(h)} = (\phi_1, \dots, \phi_p, 0, \dots, 0)^T$, and $\hat{\phi}_{hh}$ is the last component of the order-h Y-W estimator. By Theorem 3.3.1, $\sqrt{n}\,\hat{\phi}_{hh} \xrightarrow{d} N(0, \sigma^2(\Gamma_h^{-1})_{hh})$.

Step 2. $1/(\Gamma_h^{-1})_{hh} = \mathrm{Var}(X_{t-h} - P(X_{t-h} \mid X_{t-h+1}, \ldots, X_{t-1}))$, which by time reversal equals the one-step error of X_t from $X_{t-1}, \ldots, X_{t-h+1}$, i.e. σ^2 as $h - 1 \geq p$ (Proposition 2.5.B, Step 2). So $\sigma^2(\Gamma_h^{-1})_{hh} = 1$. ■

So $\pm 1.96/\sqrt{n}$ SPACF bands beyond lag p choose the AR order.

3.4 Least squares estimators

Motivation. Regressing X_t on its lags gives the same limit law as Yule–Walker despite dependent regressors.

For AR(1) $X_t = \phi X_{t-1} + \epsilon_t$, the *least squares estimator* minimizes $S(\phi) = \sum_{t=2}^n (X_t - \phi X_{t-1})^2$:

$$\hat{\phi} = \frac{\sum_{t=2}^n X_t X_{t-1}}{\sum_{t=2}^n X_{t-1}^2}, \qquad \sqrt{n}(\hat{\phi} - \phi) = \frac{n^{-1/2} \sum_{t=2}^n X_{t-1}\epsilon_t}{n^{-1} \sum_{t=2}^n X_{t-1}^2}.$$

Here $EX_{t-1}\epsilon_t = 0$, $\mathrm{Var}(\sum_{t=2}^n X_{t-1}\epsilon_t) = (n-1)\sigma^2 EX_1^2$ (cross terms vanish as $\epsilon_t \perp X_{s-1}\epsilon_s X_{t-1}$, $s < t$), and

$$\frac{1}{\sqrt{n}} \sum_{t=2}^n X_{t-1}\epsilon_t \xrightarrow{d} N(0, \sigma^2 EX_1^2) \tag{3.3}$$

(Lemma 3.3.A(b)), while the denominator tends to $\gamma(0) = EX_1^2$.

Theorem 3.4.1 (LS estimator in AR(1))

If $\hat{\phi}$ is the least squares estimator of ϕ in a stationary AR(1) model, then

$$\sqrt{n}(\hat{\phi} - \phi) \xrightarrow{d} N(0, (EX_1^2)^{-1}\sigma^2) = N(0, 1 - \phi^2).$$

Proof. [Supplement] Numerator $\xrightarrow{d} N(0, \sigma^2 EX_1^2)$ by (3.3), denominator $\to EX_1^2 > 0$ a.s. (Lemma 3.3.A(a)); Slutsky gives $N(0, \sigma^2/EX_1^2)$, and $EX_1^2 = \sigma^2/(1-\phi^2)$ (Example 2.2.3). (Case $p = 1$ of Theorem 3.4.2.) ■

For AR(p) with $\mathbf{X}_{t-1} = (X_{t-1}, \ldots, X_{t-p})^T$, the LS estimator is

$$\hat{\boldsymbol{\phi}} = (\hat{\phi}_1, \ldots, \hat{\phi}_p)^T = \Big(\sum_{t=p+1}^n \mathbf{X}_{t-1}\mathbf{X}_{t-1}^T \Big)^{-1} \sum_{t=p+1}^n \mathbf{X}_{t-1} X_t,$$

so that, using $X_t = \mathbf{X}_{t-1}^T \boldsymbol{\phi} + \epsilon_t$,

$$\hat{\boldsymbol{\phi}} - \boldsymbol{\phi} = \Big(\sum_{t=p+1}^n \mathbf{X}_{t-1}\mathbf{X}_{t-1}^T \Big)^{-1} \sum_{t=p+1}^n \mathbf{X}_{t-1}\epsilon_t.$$

Here $E\mathbf{X}_{t-1}\epsilon_t = 0$ and $\mathrm{Cov}(\sum_{t=p+1}^n \mathbf{X}_{t-1}\epsilon_t) = (n-p)\Gamma\sigma^2$ with $\Gamma_{ij} = \gamma(i-j)$.

Theorem 3.4.2 (LS estimator in AR(p))

If $\hat{\boldsymbol{\phi}}$ is the least squares estimator of $\boldsymbol{\phi}$ in a stationary (causal) AR(p) model, then

$$\sqrt{n}(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi}) \xrightarrow{d} N(0, \Gamma^{-1}\sigma^2).$$

Proof. [Supplement]

Step 1. By Lemma 3.3.A(b), $n^{-1/2}\sum_{t=p+1}^{n}\mathbf{X}_{t-1}\epsilon_t \xrightarrow{d} N(0,\Gamma\sigma^2)$.

Step 2. By Lemma 3.3.A(a),(c), $n^{-1}\sum_{t=p+1}^{n}\mathbf{X}_{t-1}\mathbf{X}_{t-1}^T \to \Gamma$ a.s. with Γ nonsingular, so the inverse exists for large n and converges to Γ^{-1}.

Step 3. Slutsky: $\sqrt{n}(\hat{\boldsymbol{\phi}}-\boldsymbol{\phi}) \xrightarrow{d} \Gamma^{-1}N(0,\Gamma\sigma^2) = N(0,\Gamma^{-1}\sigma^2)$. ■

So Y-W and LS have the same limit law (Theorems 3.3.1, 3.4.2). With $\hat{\Gamma}_{ij}=\hat{\gamma}(i-j)$ and consistent $\hat{\sigma}^2$,

$$\frac{n(\hat{\boldsymbol{\phi}}-\boldsymbol{\phi})^T\hat{\Gamma}(\hat{\boldsymbol{\phi}}-\boldsymbol{\phi})}{\hat{\sigma}^2} \xrightarrow{d} \chi^2(p), \tag{3.4}$$

since $Z_n=\sqrt{n}\,\Gamma^{1/2}(\hat{\boldsymbol{\phi}}-\boldsymbol{\phi})/\sigma \xrightarrow{d} N(0,I_p)$ and the statistic is $Z_n^TZ_n+o_p(1)$. A $100(1-\alpha)\%$ confidence region is

$$S_\alpha=\Big\{\boldsymbol{\phi}\in\mathbb{R}^p : \frac{n(\hat{\boldsymbol{\phi}}-\boldsymbol{\phi})^T\hat{\Gamma}(\hat{\boldsymbol{\phi}}-\boldsymbol{\phi})}{\hat{\sigma}^2} \le \chi^2(p,1-\alpha)\Big\},$$

where $P(\chi^2(p)\ge\chi^2(p,1-\alpha))=\alpha$.

3.5 Maximum likelihood estimators

Motivation. The Gaussian AR likelihood factorizes into one-step conditionals; dropping the first factor gives the conditional MLE, which is least squares.

Assume $\{\epsilon_t\}\sim IID\,N(0,\sigma^2)$ (the usual working assumption even for non-normal data).

3.5.1 Exact likelihood of AR(1)

Let $X_t=\phi X_{t-1}+\epsilon_t$, $\{\epsilon_t\}\sim IID\,N(0,\sigma^2)$, $|\phi|<1$. Then $EX_1=0$, $\mathrm{Var}(X_1)=\sigma^2/(1-\phi^2)$, so

$$f_{X_1}(x_1;\phi,\sigma^2)=\frac{1}{\sqrt{2\pi\sigma^2/(1-\phi^2)}}\exp\Big\{\frac{-x_1^2}{2\sigma^2/(1-\phi^2)}\Big\}.$$

As $\epsilon_2\perp X_1$,

$$P(X_2\le x\mid X_1=x_1)=P(\phi X_1+\epsilon_2\le x\mid X_1=x_1)=P(\epsilon_2\le x-\phi x_1),$$

so $X_2\mid X_1=x_1\sim N(\phi x_1,\sigma^2)$; by the Markov structure $f_{X_t|X_{t-1},\dots,X_1}=f_{X_t|X_{t-1}}$ is the $N(\phi x_{t-1},\sigma^2)$ density. Hence

$$f_{X_1,\dots,X_n}(x_1,\dots,x_n;\phi,\sigma^2)=f_{X_1}(x_1;\phi,\sigma^2)\prod_{t=2}^{n}f_{X_t|X_{t-1}}(x_t\mid x_{t-1};\phi,\sigma^2), \tag{3.5}$$

and the log-likelihood is

$$\begin{aligned}L(\phi,\sigma^2)=&-\frac{1}{2}\log(2\pi)-\frac{1}{2}\log\frac{\sigma^2}{1-\phi^2}-\frac{x_1^2}{2\sigma^2/(1-\phi^2)}\\&-\frac{n-1}{2}\log(2\pi\sigma^2)-\frac{1}{2\sigma^2}\sum_{t=2}^{n}(x_t-\phi x_{t-1})^2.\end{aligned} \tag{3.6}$$

The score equations are nonlinear in ϕ, so (3.6) is maximized numerically. For AR(p) the first factor is the $N(0,\Gamma_p)$ density of $(X_1,\dots,X_p)$; ARMA MLEs are also numerical.

3.5.2 Conditional likelihood

The *conditional maximum likelihood estimator* drops f_{X_1} from (3.5):

$$L_c(\phi, \sigma^2) = -\frac{n-1}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{t=2}^{n}(x_t - \phi x_{t-1})^2.$$

Its maximizers are

$$\hat{\phi} = \frac{\sum_{t=2}^{n} X_t X_{t-1}}{\sum_{t=2}^{n} X_{t-1}^2}, \qquad \hat{\sigma}^2 = \frac{1}{n-1}\sum_{t=2}^{n}(X_t - \hat{\phi}X_{t-1})^2.$$

Proposition 3.5.A (Conditional MLE = LS for AR(p); supplement)

For a Gaussian causal AR(p), the conditional MLE of $\boldsymbol{\phi}$ given $X_1, \dots, X_p$ equals the LS estimator $\hat{\boldsymbol{\phi}}$, the conditional MLE of σ^2 is $\hat{\sigma}^2 = (n-p)^{-1}\sum_{t=p+1}^{n}(X_t - \mathbf{X}_{t-1}^T\hat{\boldsymbol{\phi}})^2 \to \sigma^2$ a.s., and $\sqrt{n}(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi}) \xrightarrow{d} N(0, \Gamma^{-1}\sigma^2)$. The exact MLE has the same limit distribution.

Proof. [Supplement]

Step 1. $L_c = -\frac{n-p}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}S(\boldsymbol{\phi})$, $S(\boldsymbol{\phi}) = \sum_{t=p+1}^{n}(X_t - \mathbf{X}_{t-1}^T\boldsymbol{\phi})^2$, is maximized in $\boldsymbol{\phi}$ by the LS estimator for every σ^2; $\partial L_c/\partial\sigma^2 = -\frac{n-p}{2\sigma^2} + \frac{S(\hat{\boldsymbol{\phi}})}{2\sigma^4} = 0$ gives $\hat{\sigma}^2 = S(\hat{\boldsymbol{\phi}})/(n-p)$.

Step 2. Writing $X_t - \mathbf{X}_{t-1}^T\hat{\boldsymbol{\phi}} = \epsilon_t - \mathbf{X}_{t-1}^T(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi})$ and using the normal equations, $S(\hat{\boldsymbol{\phi}})/(n-p) = n^{-1}\sum\epsilon_t^2 - (\hat{\boldsymbol{\phi}} - \boldsymbol{\phi})^T(n^{-1}\sum\mathbf{X}_{t-1}\mathbf{X}_{t-1}^T)(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi}) + o(1) \to \sigma^2$ (SLLN, Lemma 3.3.A, Theorem 3.4.2); the law of $\hat{\boldsymbol{\phi}}$ is Theorem 3.4.2.

Step 3. (Exact MLE, sketch.) The extra log-density of $(X_1, \dots, X_p)$ has $O_p(1)$ derivative while the Hessian of L_c is $\approx -n\Gamma/\sigma^2$; a one-step expansion shows the maximizers differ by $O_p(1/n)$, so the limits coincide. ■

3.6 Prediction

Motivation. The mean-square optimal forecast is a conditional expectation, computed recursively for ARMA models.

Given $X_1, \dots, X_n$, the *predictor* of X_{n+h} is

$$f(X_1, \dots, X_n) := E(X_{n+h} \mid X_1, \dots, X_n).$$

Proposition 3.6.A (Optimality of the conditional mean; supplement)

If $EX_{n+h}^2 < \infty$, then $E(X_{n+h} - g(X_1, \dots, X_n))^2$ over all measurable g with $Eg^2 < \infty$ is minimized by $g = f$.

Proof. [Supplement] With $Y = X_{n+h}$, $\mathbf{X} = (X_1, \dots, X_n)$,

$$E(Y - g(\mathbf{X}))^2 = E(Y - f(\mathbf{X}))^2 + E(f(\mathbf{X}) - g(\mathbf{X}))^2 + 2E\big[(Y - f(\mathbf{X}))(f(\mathbf{X}) - g(\mathbf{X}))\big],$$

and the cross term is $2E\big[(f-g)(\mathbf{X})\,E(Y - f(\mathbf{X}) \mid \mathbf{X})\big] = 0$. So $E(Y-g)^2 \geq E(Y-f)^2$, with equality iff $g(\mathbf{X}) = f(\mathbf{X})$ a.s. ■

For Gaussian $\{X_t\}$, $f = \sum_{i=1}^{n} a_i X_i$ (Section 1.2) with a_i minimizing $E(X_{n+h} - \sum_i a_i X_i)^2$. In general f is nonlinear, but linear predictors are used in practice (Durbin–Levinson algorithm; for ARMA, functions of the parameters).

From the infinite past, $\hat{X}_t(h) = E(X_{t+h} \mid X_t, X_{t-1}, \dots)$.

Example 3.6.1 (Forecasting an AR(1) with mean)

Let $(1-\phi B)(X_t-\mu)=\epsilon_t$, $\{\epsilon_t\}\sim IID(0,\sigma^2)$, $|\phi|<1$, so $X_t=\mu+\phi(X_{t-1}-\mu)+\epsilon_t$. As $\epsilon_{t+1}\perp X_t, X_{t-1},\dots$,

$$\hat{X}_t(1)=E(\mu+\phi(X_t-\mu)+\epsilon_{t+1}\mid X_t,X_{t-1},\dots)=\mu+\phi(X_t-\mu).$$

Similarly $\hat{X}_t(h)=\mu+\phi(\hat{X}_t(h-1)-\mu)$, hence

$$\hat{X}_t(h)=\mu+\phi^h(X_t-\mu)\longrightarrow\mu\quad(h\to\infty).$$

In practice μ,ϕ are replaced by $\hat{\mu},\hat{\phi}$.

For a stationary invertible ARMA(1,1), $(1-\phi B)(X_t-\mu)=(1+\theta B)\epsilon_t$, $X_{t+1}=\mu(1-\phi)+\phi X_t+\epsilon_{t+1}+\theta\epsilon_t$, and by invertibility $E(\epsilon_t\mid X_t,X_{t-1},\dots)=\epsilon_t$:

$$\hat{X}_t(1)=\mu(1-\phi)+\phi X_t+\theta\epsilon_t,\qquad \hat{X}_t(h)=\mu(1-\phi)+\phi\hat{X}_t(h-1),\quad h\geq 2.$$

In practice ϵ_t is replaced by the residual $\hat{\epsilon}_t$ and μ,ϕ,θ by estimates.

3.7 Selected exercises

Motivation. Exercises on the mean interval, hand computation of sample ACF/PACF, and $|\hat{\rho}(h)|\leq 1$.

Exercise 3.1 (Inference on the mean of an AR(1))

In an AR(1) model $X_t-\mu=\phi(X_{t-1}-\mu)+\epsilon_t$ with $\phi=0.3$ and $\sigma^2=1$, a sample of size $n=100$ gives $\bar{X}=0.1$.
(a) Find an approximate 95% confidence interval for μ.
(b) Test $H_0:\mu=0$ against $H_1:\mu\neq 0$ at level 5%.

Solution.

(a) By Theorem 3.1.1, $\sqrt{n}(\bar{X}-\mu)(1-\phi)/\sigma\xrightarrow{d}N(0,1)$, so $P(|\bar{X}-\mu|\leq z_{0.025}\sigma/(\sqrt{n}(1-\phi)))\to 0.95$. Here

$$\frac{z_{0.025}\,\sigma}{\sqrt{n}(1-\phi)}=\frac{1.96\times 1}{10\times 0.7}=0.28,$$

so the interval is $0.1\pm 0.28=[-0.18,\,0.38]$.

(b) Reject when $|\sqrt{n}\bar{X}|\geq z_{0.025}\sigma/(1-\phi)=1.96/0.7=2.8$; here $|\sqrt{n}\bar{X}|=1<2.8$, so H_0 is not rejected.

An i.i.d. treatment with $\gamma(0)=1/0.91$ gives half-width $1.96\times(0.91)^{-1/2}/10\approx 0.205$, too narrow. ■

Exercise 3.2 (Sample ACVF, ACF and PACF by hand)

For the data

$$(x_1,\dots,x_{10})=(16,\ 9,\ 10,\ 3,\ 5,\ 12,\ 14,\ 7,\ 9,\ 11),$$

compute $\hat{\gamma}(h)$ and $\hat{\rho}(h)$ for $h=0,1,2,3$, and $\hat{\phi}_{hh}$ for $h=1,2,3$.

Solution.

Step 1. $\bar{x} = 96/10 = 9.6$, and the deviations $d_t = x_t - \bar{x}$ are

$$6.4,\ -0.6,\ 0.4,\ -6.6,\ -4.6,\ 2.4,\ 4.4,\ -2.6,\ -0.6,\ 1.4.$$

Step 2. By Definition 3.2.1, $\hat{\gamma}(h) = \frac{1}{10}\sum_{t=1}^{10-h} d_t d_{t+h}$. The lagged cross-product sums are 140.40, 12.44, −43.92, −50.88 for $h = 0, 1, 2, 3$, so

$$\hat{\gamma}(0) = 14.04, \quad \hat{\gamma}(1) = 1.244, \quad \hat{\gamma}(2) = -4.392, \quad \hat{\gamma}(3) = -5.088,$$

and, dividing by $\hat{\gamma}(0)$,

$$\hat{\rho}(0) = 1, \quad \hat{\rho}(1) = 0.0886, \quad \hat{\rho}(2) = -0.3128, \quad \hat{\rho}(3) = -0.3624.$$

Step 3. By Definition 3.2.3, $\hat{\phi}_{11} = \hat{\rho}(1) = 0.0886$, and solving the 2×2 system as in Example 2.5.2,

$$\hat{\phi}_{22} = \frac{\hat{\rho}(2) - \hat{\rho}(1)^2}{1 - \hat{\rho}(1)^2} = \frac{-0.3128 - 0.0079}{1 - 0.0079} = -0.3232.$$

Step 4. For $h = 3$, Cramer's rule gives $\hat{\phi}_{33} = \det \hat{R}_3^* / \det \hat{R}_3$ with $\hat{R}_3 = [\hat{\rho}(i-j)]_{i,j=1}^3$ and $\hat{R}_3^*$ its last column replaced by $(\hat{\rho}(1), \hat{\rho}(2), \hat{\rho}(3))^T$. Numerically $\det \hat{R}_3 = 0.8815$ and $\det \hat{R}_3^* = -0.2947$, so $\hat{\phi}_{33} = -0.3344$.

All values lie within $\pm 1.96/\sqrt{10} = \pm 0.62$ (Corollary 3.2.A): no significant autocorrelation. ■

Exercise 3.4 (Sample autocorrelations are bounded by one)

Let $x_1, \ldots, x_n$ be not all equal.
(a) Show that $|\hat{\rho}(h)| \le 1$ for all $|h| < n$.
(b) Show that the matrix $\hat{\Gamma}_k = [\hat{\gamma}(i-j)]_{i,j=1}^k$ is nonnegative definite for every $k \ge 1$.

Solution. Let $y_t = x_t - \bar{x}$ for $1 \le t \le n$ and $y_t = 0$ for all other integers t. Since $\hat{\gamma}(h) = \hat{\gamma}(-h)$ and the extra terms vanish,

$$\hat{\gamma}(h) = \frac{1}{n}\sum_{t \in \mathbb{Z}} y_t y_{t+h}, \qquad h \in \mathbb{Z}.$$

(a)

Step 1. $\hat{\gamma}(0) = n^{-1}\sum_t y_t^2 > 0$ because the x_t are not all equal, so $\hat{\rho}(h)$ is defined.

Step 2. By the Cauchy–Schwarz inequality for finite sums,

$$\Big|\sum_t y_t y_{t+h}\Big| \le \Big(\sum_t y_t^2\Big)^{1/2}\Big(\sum_t y_{t+h}^2\Big)^{1/2} = \sum_t y_t^2,$$

since shifting the index does not change $\sum_t y_t^2$. Dividing by $n\hat{\gamma}(0)$ gives $|\hat{\rho}(h)| \le 1$.

(b) For $a \in \mathbb{R}^k$, substituting $s = t - j$ in the inner sum (all sums are finite),

$$\sum_{i,j=1}^k a_i a_j \hat{\gamma}(i-j) = \frac{1}{n}\sum_{i,j} a_i a_j \sum_s y_{s+j} y_{s+i} = \frac{1}{n}\sum_{s \in \mathbb{Z}}\Big(\sum_{i=1}^k a_i y_{s+i}\Big)^2 \ge 0.$$

(b) also implies (a) via the 2×2 minor $\hat{\gamma}(0)^2 - \hat{\gamma}(h)^2 \ge 0$; zero-padding is why the divisor is n. ■